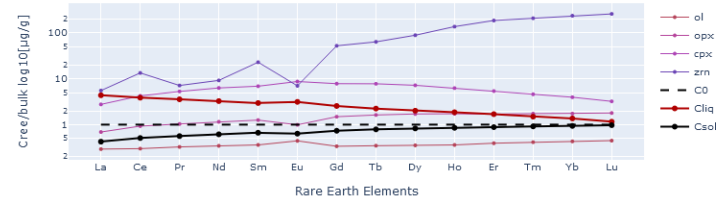
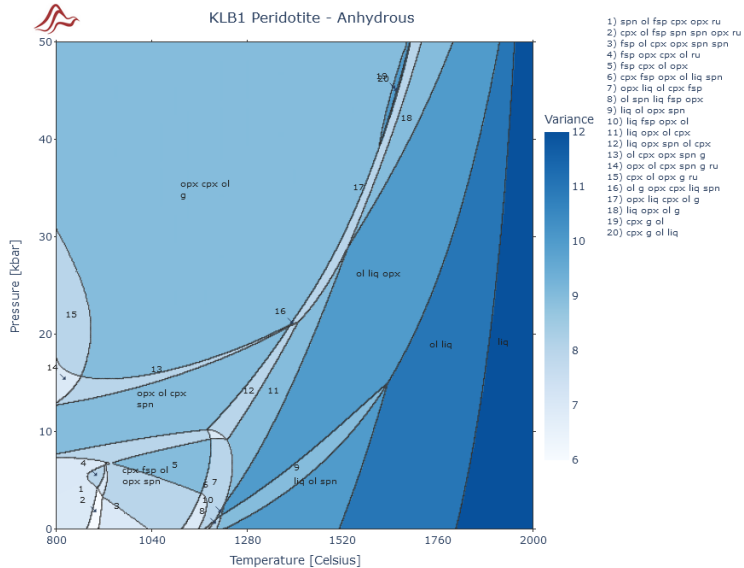


Phase Equilibrium Modelling with MAGeMin

Nicolas Riel - Johannes-Gutenberg University Mainz, Germany nriel@uni-mainz.de



Overview

- General introduction and rationales
- Benchmarks and MAGEMin packages
- MAGEMinApp
- MAGEMin_C



<https://github.com/ComputationalThermodynamics/Resources/GG2024>

<https://github.com/ComputationalThermodynamics/Resources/Mainz2025>

Why a new Gibbs free energy minimizer?

Perple_**x**



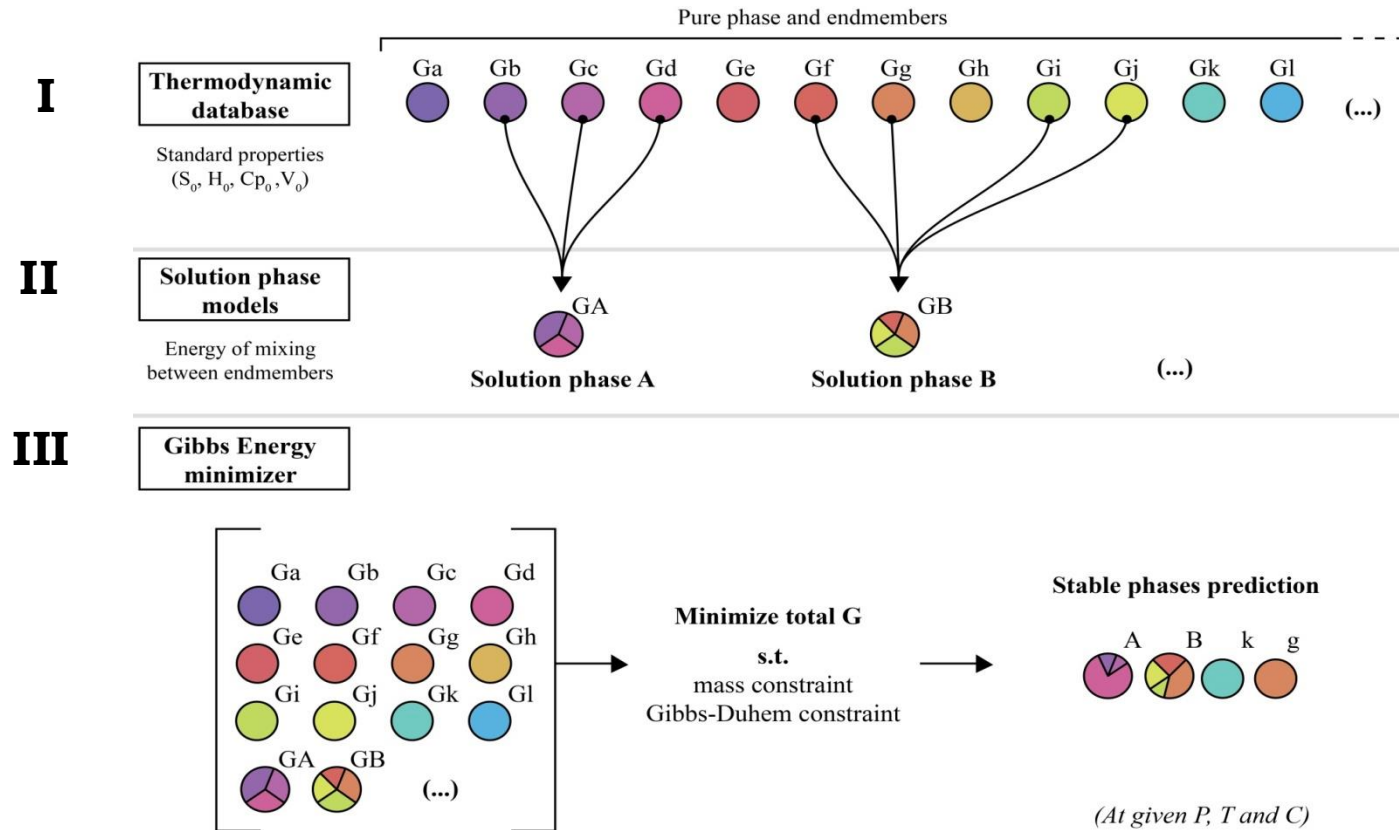
MELTS

- Developed (in the 90's) to produce phase diagrams
- Natively not parallel
- Minimizations can often fail
- Not optimized for solution update
- Difficult to couple to geodynamic codes (written in Fortran)

Constraints for a new minimizer

- Open source and code efficient
 - Github + C language
- Built-in database
 - No input files*
- Fully parallel for scalability
 - MPI & multi-threading
- Easily callable by external software
 - Julia wrapper
- Fast and stable
 - Efficient optimization methods
- Use most recent/complex thermodynamic database
 - e.g., Holland et al., 2018
 - Weller et al., *submitted*
(igneous database upgrade)

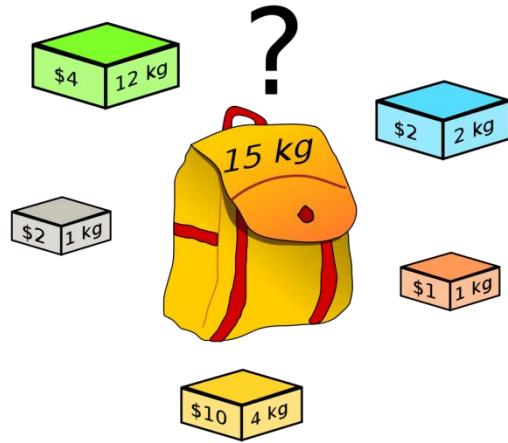
How to compute a stable phase equilibrium?



e.g.,
 tc-ds6xx.txt for TC
 hp02ver.dat for Perple_X

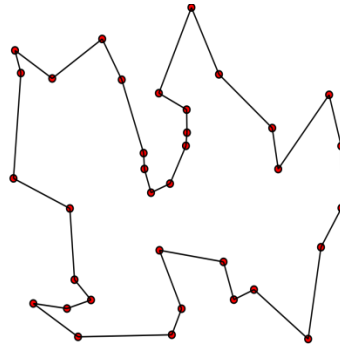
Examples of optimization problems

knapsack



Maximize value of backpack
→ Weight must sum to 15kg

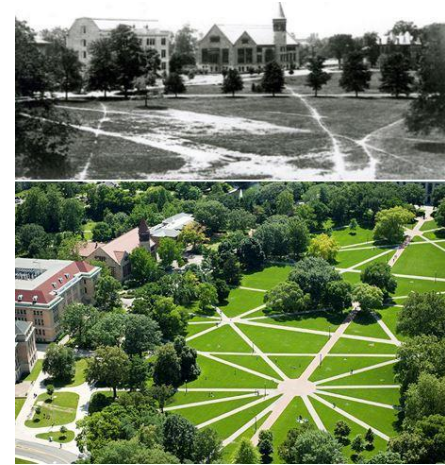
Travelling salesman



Minimize path length
→ Must connect every city

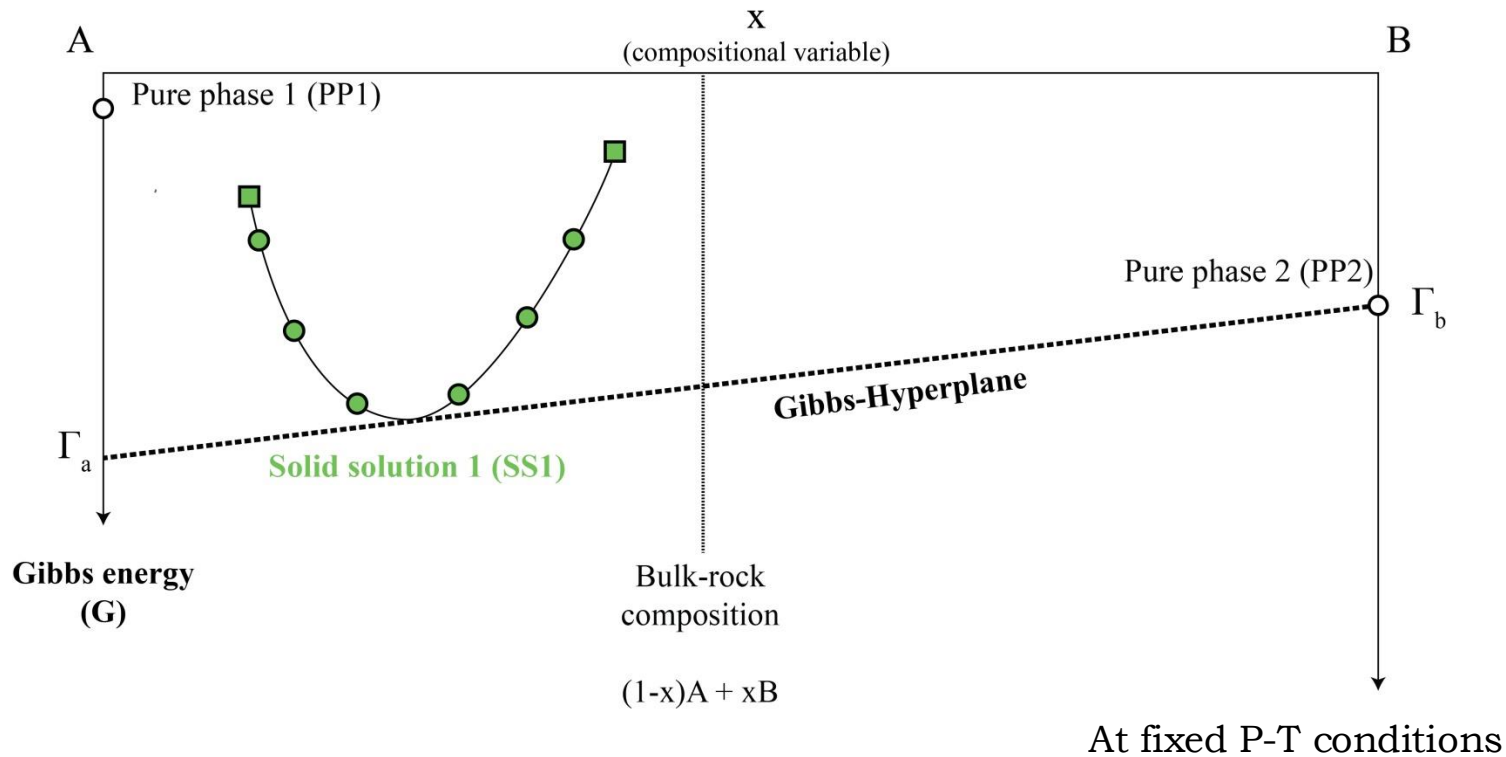
Walk path

Ohio State University

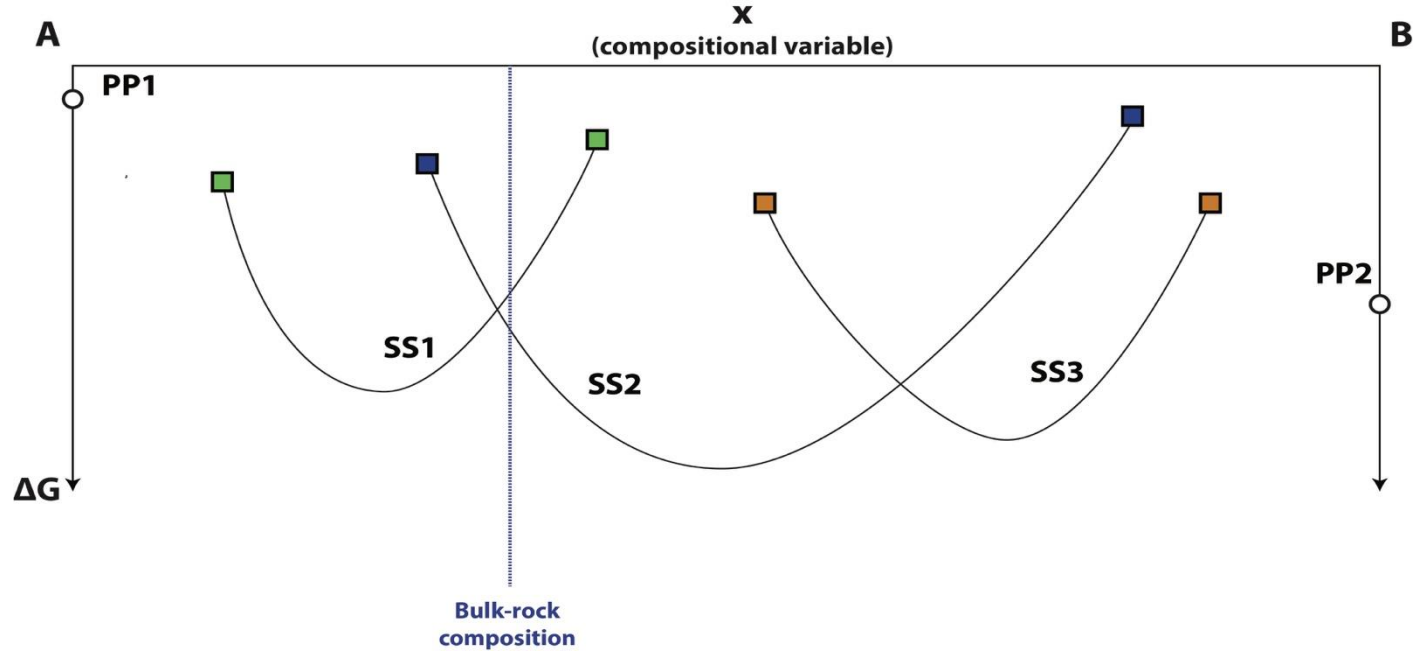


Minimize distance between points
to connect
→ Saves on grass seed

Phase equilibria: introduction

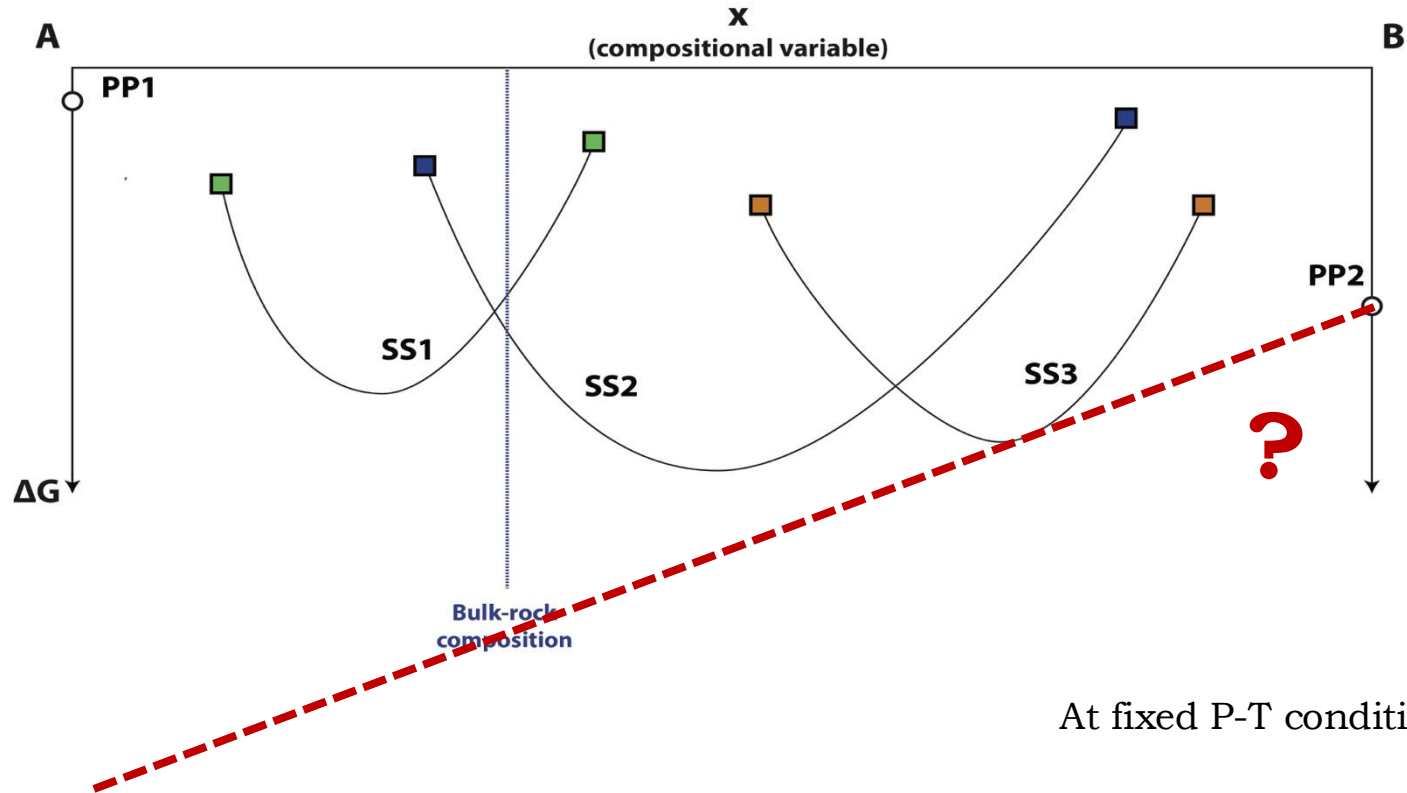


Phase equilibria: introduction

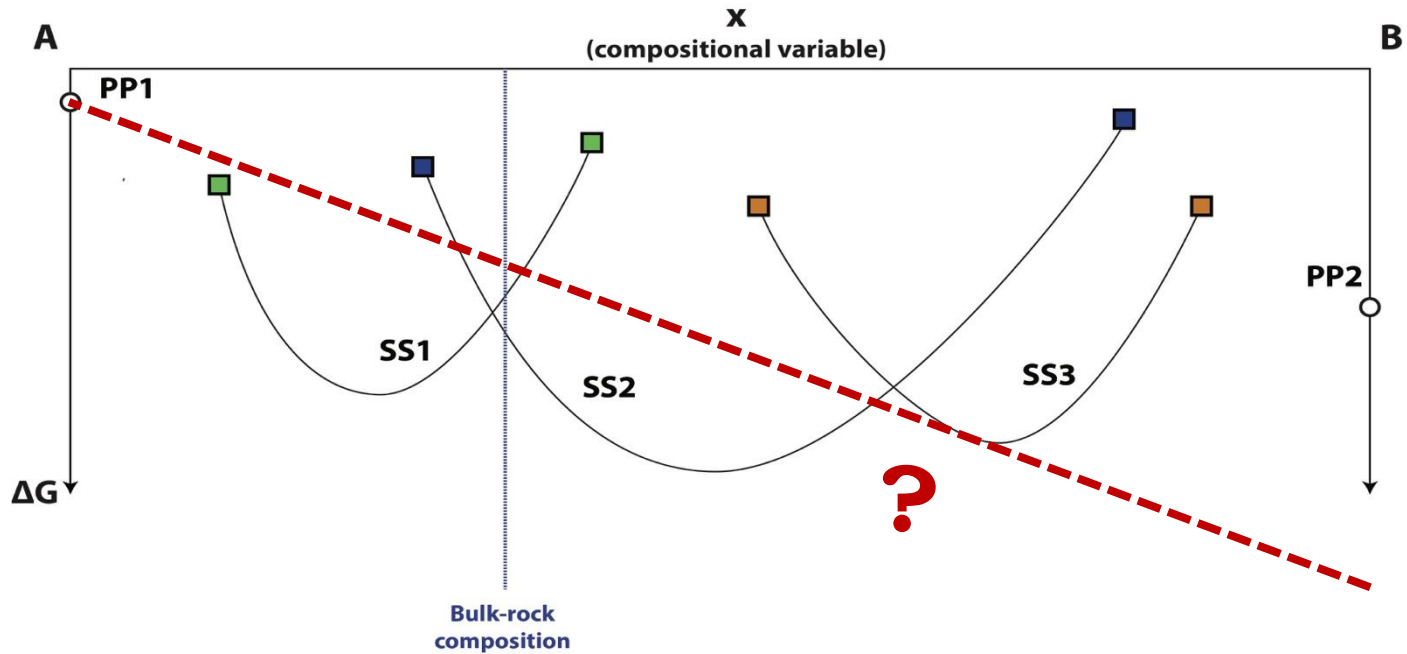


At fixed P-T conditions

Phase equilibria: introduction

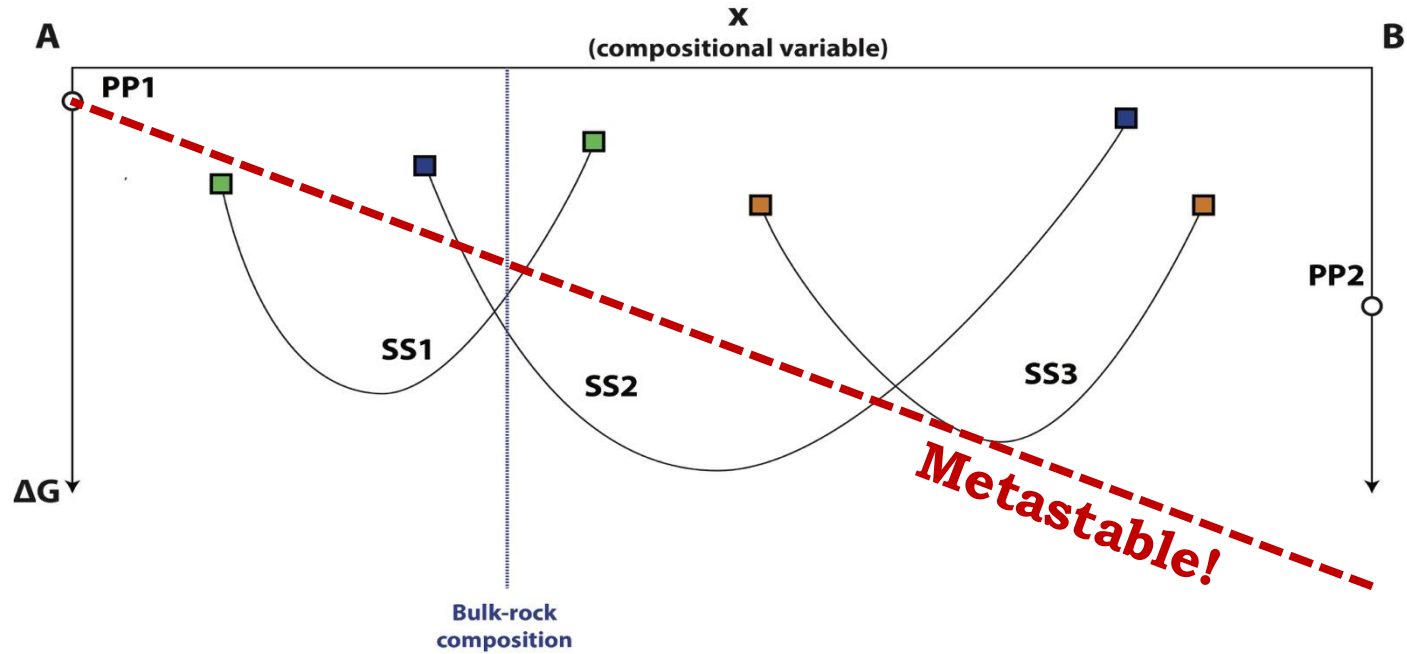


Phase equilibria: introduction



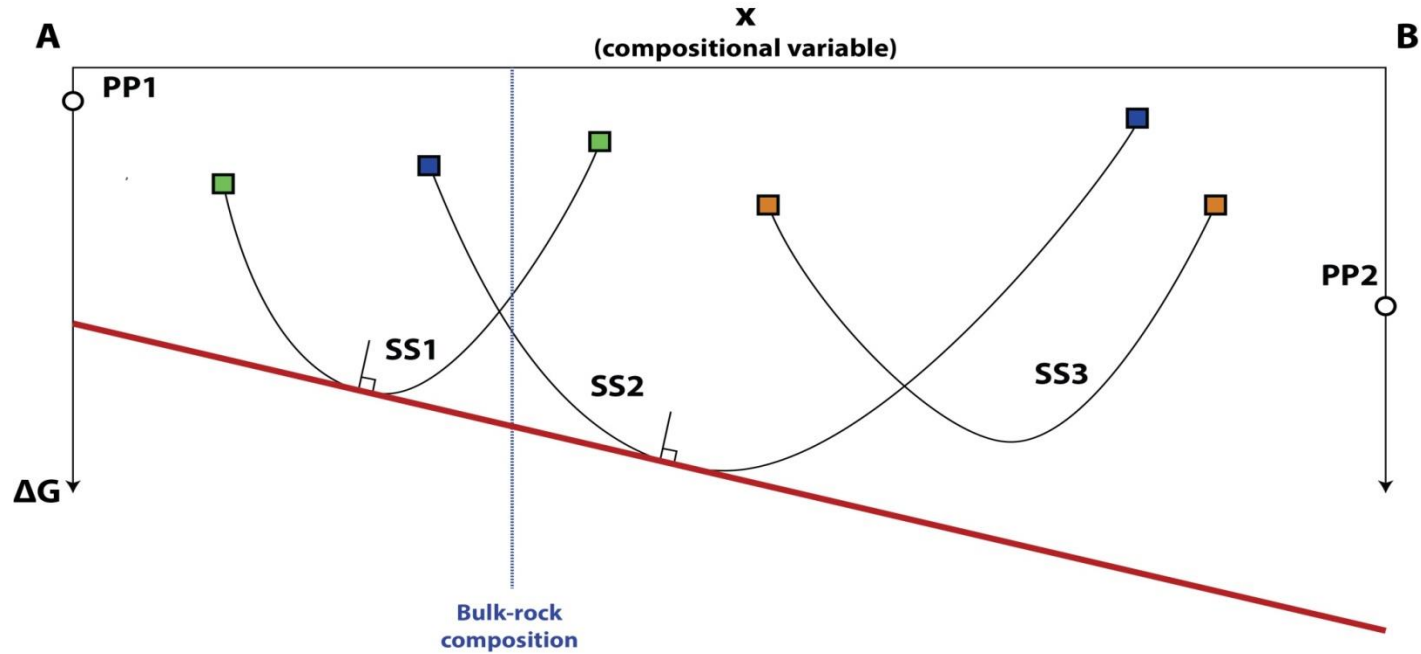
At fixed P-T conditions

Phase equilibria: introduction



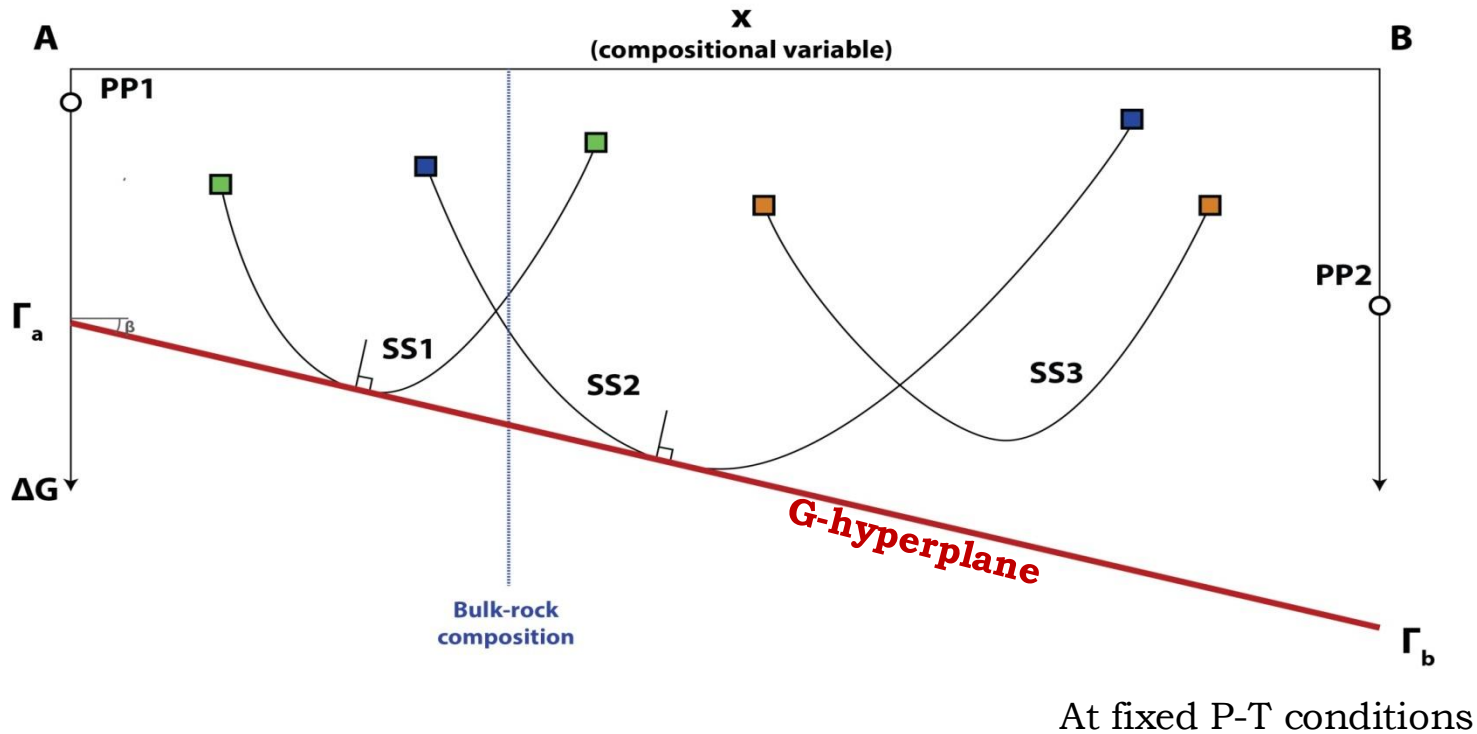
At fixed P-T conditions

Phase equilibria: introduction

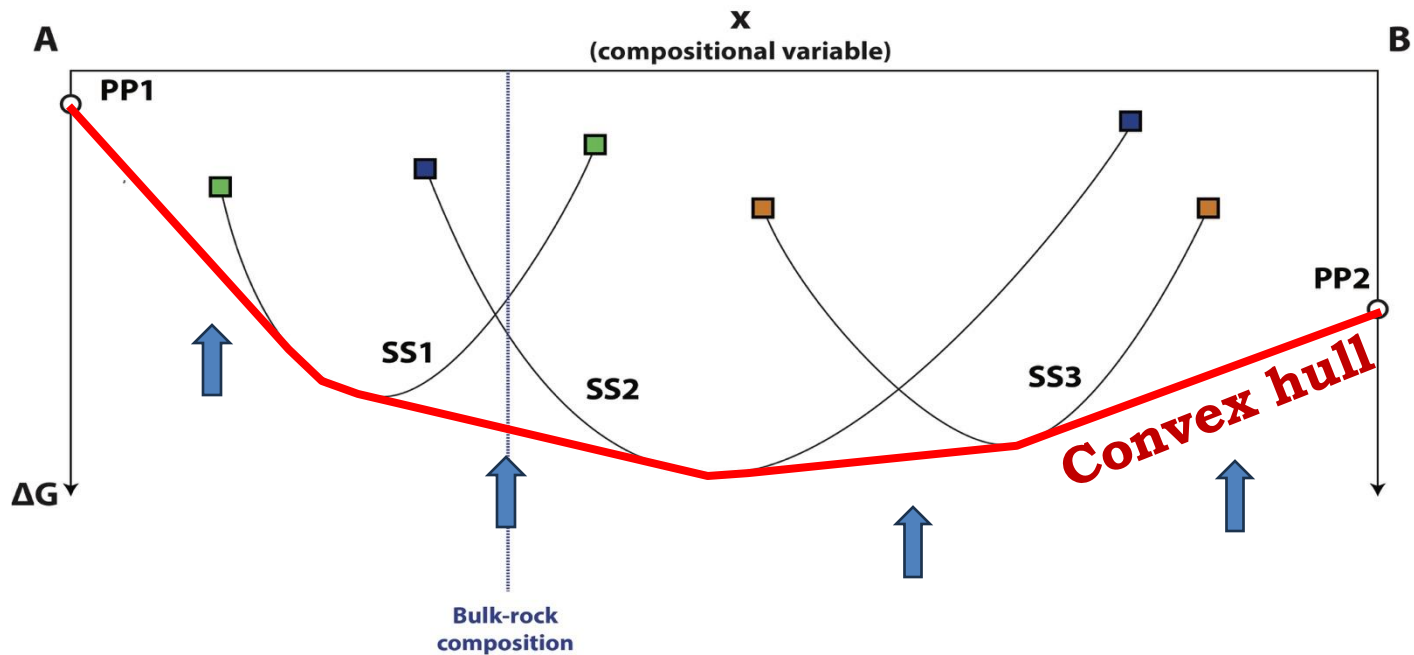


At fixed P-T conditions

Phase equilibria: introduction



Phase equilibria: introduction



At fixed P-T conditions

Gibbs free energy minimization: a challenging optimization problem

- Given a set of minerals at given P , T and C

Minimize
(at fixed P, T, C)

$$G_{\text{sys}} = \underbrace{\sum_{\lambda=1}^{\Lambda} \alpha_{\lambda} \sum_{i=1}^{N_{\lambda}} \mu_{i(\lambda)} p_{i(\lambda)}}_{\substack{\text{Solution phase} \\ \text{(e.g. olivine)}}} + \underbrace{\sum_{\omega=1}^{\Omega} \alpha_{\omega} \mu_{\omega}}_{\substack{\text{Pure phase} \\ \text{(e.g. quartz)}}$$

where

$$\mu_{i(\lambda)} = g_{i(\lambda)}^0 + RT \log(\text{idm}_{i(\lambda)}) + g_{i(\lambda)}^{\text{ex}}$$

$$\text{s.t. } \text{idm}_i > 0.0$$

s.t.
Mass constraint

$$\sum_{\lambda=1}^{\Lambda} \alpha_{\lambda} \sum_{i=1}^{N_{\lambda}} a_{ij} p_{i(\lambda)} + \sum_{\omega=1}^{\Omega} \alpha_{\omega} a_{\omega j} - b_j = 0,$$

And Gibbs-Duhem
constraint

$$\mu_{i(\lambda), \omega} = \sum_{j=1}^C a_{i, \omega j} \Gamma_j,$$

MAGEMin framework

(Mineral Assemblage Gibbs Energy Minimization)

MAGEMin



- MPI-parallel C code
- Point-wise minimization at given P-T-X
- Metapelite (White et al., 2014)
- Metabasite (Green et al., 2016)
- Igneous (Holland et al., 2018)
- Ultramafic (Evans & Forst, 2021)
- Mantle (Stixude & Lithgow-Bertelloni, 2010)

MAGEMin_C



- Julia wrapper of the C code
- Flexible programming interface
- Database calibration
- Geodynamic coupling

MAGEMinApp



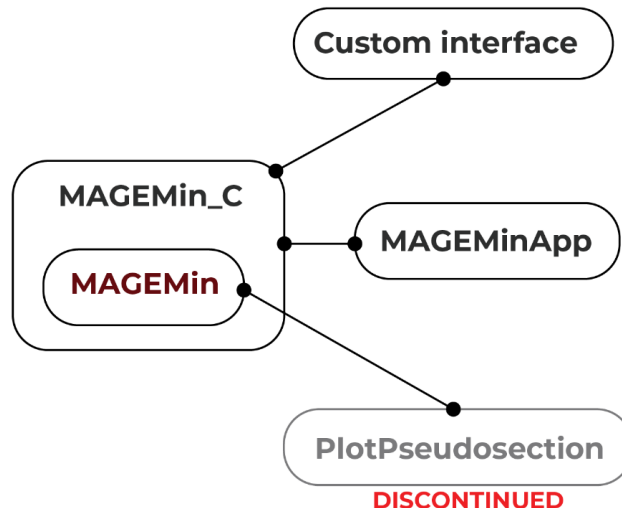
- Web browser app (graphic user interface)
- Parallel point-wise minimization
- PT, PX, TX and PT-X phase diagrams
- Auto labelling, contouring
- Fractional melting/crystallization paths

PlotPseudosection



- Matlab app (Graphic user interface)
- Parallel point-wise minimization
- PT, PX, TX diagrams
- Contouring
- PT paths
- Trace element partitioning for mafic to ultramafic systems

E.G., GEODYNAMIC COUPLING



MAGEMin framework

(Mineral Assemblage Gibbs Energy Minimization)

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MAGEMinApp



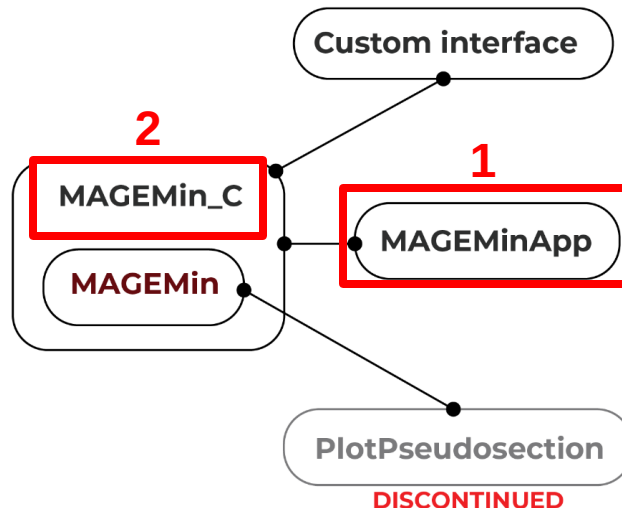
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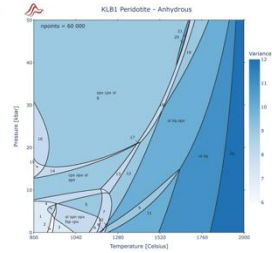


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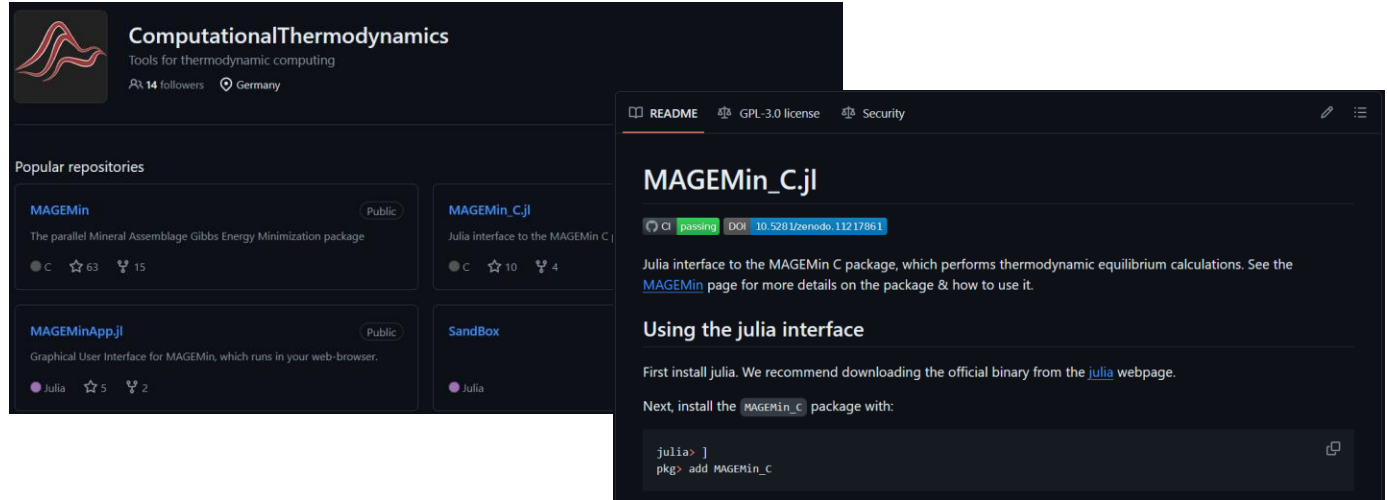
E.G., GEODYNAMIC COUPLING



MAGEMin Github: ComputationalThermodynamics



nriel@uni-mainz.de



<https://github.com/ComputationalThermodynamics>

https://github.com/ComputationalThermodynamics/MAGEMin_C.jl

<https://github.com/ComputationalThermodynamics/MAGEMinApp.jl>

<https://github.com/ComputationalThermodynamics/Resources/GG2024>

- **Do not hesitate to contact me if anything is wrong/broken or needs to be added**
I cannot correct bugs or add new options if I am unaware of them!
- **Everything is open source: you can contribute by creating pull-requests on Github!**